tutorial

(1) Describe quantum teleportation where the pre-shared entangled state is $|\psi^-\rangle$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$$

$|\gamma\rangle =$ Teleport $\alpha|0\rangle + \beta|1\rangle$

(2) Werner state – mixed state component + maximally entangled component

$$\hat{\rho}_w = \frac{p}{4} I_2 \otimes I_2 + (1-p)|\psi^-\rangle\langle\psi^-|$$

What's the density matrix for this state?

What's the purity? Purity $= Tr(\rho^2)$

for problem set

Fidelity $= \langle\psi|\hat{\rho}|\psi\rangle$

where $|\psi\rangle$ is the desired state.

$$\hat{\rho}_w = \begin{bmatrix} \frac{p}{4} & 0 & 0 & 0 \\ 0 & \frac{2-p}{4} & \frac{p-1}{2} & 0 \\ 0 & \frac{p-1}{2} & \frac{2-p}{4} & 0 \\ 0 & 0 & 0 & \frac{p}{4} \end{bmatrix}$$

Purity $= \frac{3}{4}p^2 - \frac{3}{2}p + 1$